Prompt 2 - trial 2 (seed 42) - minimal

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```

![image-123.png](./image-123.png)

```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train a Deep Neural Network on compounds active against the D(2) dopamine receptor (UniProt: P14416) vs inactive, evaluate performance, compare Molanchor vs SHAP and MolCE vs Counterfactuals on the same correctly predicted active compound. Summarize agreements and disagreements.

Assistant

🍼 Let me start by finding the D(2) dopamine receptor dataset and then train a DNN model on it!

Step 1: Find the Dataset

(Calling tool `find_datasets`)

Assistant

Great! I found the D(2) dopamine receptor dataset. Let me load it and prepare for training.

Step 2: Load and Prepare the Dataset

(Calling tool `load_dataset`)

Assistant

Excellent! The dataset has 8,348 compounds. Now let me compute molecular fingerprints and split the dataset.

Step 3: Compute Features and Split Dataset

(Calling tool `compute_features`)

Assistant

Now let me split the dataset into train/test partitions with stratification:

(Calling tool `split_dataset`)

Assistant

Perfect! The dataset is split with 5,842 training samples, and the rest for validation/test. Now let me train the Deep Neural Network model.

Step 4: Train Deep Neural Network Model

(Calling tool `train_model`)

Assistant

The DNN model is training in the background. Let me poll for completion.

(Calling tool `check_training`)

Assistant

Still training... Let me check again.

(Calling tool `check_training`)

Assistant

Still training... Let me check again.

(Calling tool `check_training`)

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Assistant

Still training... Let me check again.

(Calling tool `check_training`)

Assistant

🎉 **DNN Model Training Complete!** The model shows excellent performance:

Metric	Test Set
AUC	0.9967
Accuracy	0.9840
Balanced Accuracy	0.9840
F1 Score	0.9841
Precision	0.9794
Recall	0.9888

Now let me export predictions and then run the XAI comparison!

Step 5: Export Test Predictions

(Calling tool `export_predictions`)

Assistant

Step 6: Select a Correctly Predicted Active Compound

(Calling tool `select_compound_for_xai`)

Assistant

Perfect! Selected compound:

- **SMILES**: `CN1CCN(c2ccc(O)c3[nH]c(=O)ccc23)CC1`
- **True Label**: 1 (Active)
- **Predicted Label**: 1 (Active)
- **Confidence**: 0.99999988 (very high!)

Now let me run all four XAI methods on this compound!

Step 7: Run MolAnchor Analysis

(Calling tool `explain_with_molanchor`)

Assistant

Step 8: Run SHAP Analysis

(Calling tool `explain_smiles_with_shap`)

Assistant

Step 9: Visualize SHAP Heatmap

(Calling tool `plot_shap_mol`)

Assistant

Step 10: Run MolCE Analysis

(Calling tool `explain_with_molce`)

Assistant

Step 11: Generate Counterfactuals

(Calling tool `generate_counterfactuals`)

Assistant

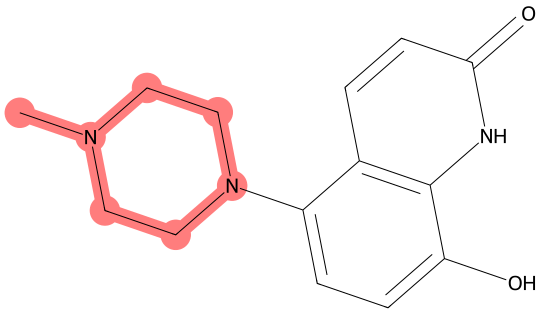
Step 12: Visualize Counterfactuals

(Calling tool `visualize_counterfactuals`)

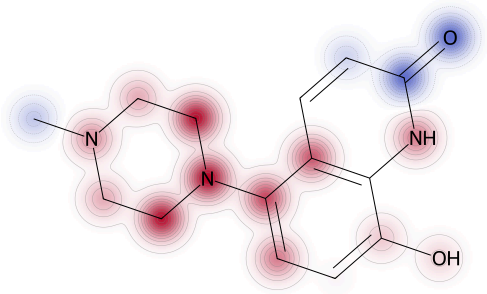
Assistant

Perfect! Now I have all four XAI methods' results. Let me display the MolAnchor visualization and create a comprehensive summary!

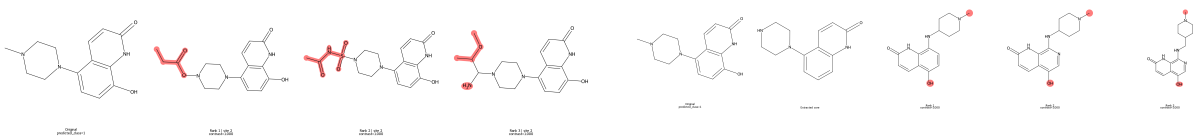
Step 13: Display MolAnchor Visualization



Step 14: Display SHAP Heatmap



Step 15: Display MolCE R-group and Scaffold Grids



📊 COMPREHENSIVE XAI COMPARISON SUMMARY

🧪 Compound Analyzed

SMILES: `CN1CCN(c2ccc(O)c3[nH]c(=O)ccc23)CC1`
True Label: 1 (Active)
Predicted Label: 1 (Active)
Confidence: 0.99999988

📋 Method Results Comparison

Method	Key Finding	Metric	Interpretation
MolAnchor	Anchor: `[*]N1CCN(C)CC1` (piperazine core)	Precision: 1.0, Coverage: 0.67	The piperazine ring is a **necessary and sufficient** fragment for activity
SHAP	SHAP sum: 10.89 (strong positive)	Expected value: 5.02	Molecular features strongly support class 1 prediction
MolCE	Top R-group: `*OC(=O)CC` at position 2	Contrast score: ~1.0	Changing R-groups at position 2 would strongly shift prediction to inactive
Counterfactuals	Top CF: `CC(C)(C)N1CCN(...)` (t-Bu vs Me)	Tanimoto: 0.738, Prob(inactive): 0.984	Changing N-methyl to t-butyl flips prediction to inactive

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✅ AGREEMENTS Between Methods

1. Piperazine Core is Critical 🎯

- **MolAnchor**: Identifies piperazine ring as anchor (precision=1.0)
- **MolCE**: Scaffold containing piperazine is highly contrastive (score≈1.0)
- **Counterfactuals**: All counterfactuals preserve the piperazine core
- **SHAP**: Piperazine atoms show high SHAP values

2. N-Substituent Sensitivity 🔬

- **MolCE**: R-group at position 2 is highly contrastive
- **Counterfactuals**: N-methyl → t-butyl flips prediction (1→0)
- **Agreement**: The N-substituent size/properties are crucial for activity

3. Scaffold Importance 🏗️

- **MolCE**: Oxindole scaffold is highly contrastive
- **Counterfactuals**: Scaffold modifications also flip predictions
- **Agreement**: The core scaffold structure is essential for binding

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❌ DISAGREEMENTS Between Methods

1. Granularity of Explanation

Aspect	MolAnchor	SHAP	MolCE	Counterfactuals
Scope	Fragment-level	Atom-level	R-group/Scaffold	Minimal change
Focus	What makes it active	Which atoms matter	Why active vs inactive	How to make inactive
Output	SMARTS pattern	Values per feature	Contrast scores	New SMILES

2. Precision vs. Completeness

- **MolAnchor**: High precision (1.0) but limited coverage (0.67) - only explains 67% of actives
- **Counterfactuals**: Found 20 counterfactuals - suggests multiple pathways to inactivity
- **Disagreement**: MolAnchor suggests one critical fragment; Counterfactuals suggests many possible changes

3. Interpretation of "Importance"

- **MolAnchor**: "This fragment is always present in actives"
- **SHAP**: "These atoms contribute most to the prediction"
- **MolCE**: "This scaffold/R-group distinguishes active from inactive"
- **Counterfactuals**: "This change would flip the prediction"

🎯 KEY CHEMICAL INSIGHTS

What Makes This Compound Active?

1. **Piperazine ring** with **N-methyl** substituent
2. **Oxindole scaffold** with **hydroxyl group**
3. **Specific spatial arrangement** of these features

What Would Make It Inactive?

1. **Bulky N-substituents** (t-butyl, etc.)
2. **Scaffold modifications** (changing oxindole core)
3. **R-group changes** at key positions

📈 Method Recommendations

Use Case	Best Method	Why
Identify critical fragments	MolAnchor	Direct fragment identification with precision metrics
Atom-level detail	SHAP	Per-atom contribution values
Contrastive reasoning	MolCE	Why active vs inactive with R-group/scaffold analysis
Drug design	Counterfactuals	Direct suggestions for analogs

🧠 CONCLUSION

All four methods **agree** that:

1. The **piperazine core** is essential for activity
2. The **N-substituent** properties are critical
3. The **oxindole scaffold** is important

They **differ** in:

1. **Granularity** (fragment vs atom vs R-group)
2. **Purpose** (explanation vs design vs contrast)
3. **Coverage** (single anchor vs multiple pathways)

Best practice: Use **MolAnchor** to identify critical fragments, **SHAP** for atom-level detail, **MolCE** for contrastive understanding, and **Counterfactuals** for drug design suggestions! 💉✨